Kaunas University of Technology
Faculty of Informatics

Machine Learning Algorithms 1

Laboratory work No. 1

Linear Regression

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Dataset: Quebec Housing Sales (target variable: sale_price)

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1. Introduction

The purpose of this laboratory work is to implement a multivariable linear regression model using two methods of coefficient estimation, the method of least squares and gradient descent, and to evaluate the resulting model. The implementation covers data preparation, calculation of the coefficients, selection of significant variables, cross-validation and testing. The results of the implemented model are compared with a model built with a standard library function (scikit-learn). Finally, the model is prepared for use with "unseen" data: a single record in the original format is transformed and its target value is predicted.

The dataset "Quebec Housing Sales" (Kaggle) is used. It contains 5,000 property sales in Quebec. The target variable is sale_price. The dataset has 12 usable features (property_id is only an identifier and is not used), i.e. fewer than 20, so all of them were kept. They include both categorical and numerical variables:

Variable	Type	Description
city	categorical (32)	City where the property is located
neighborhood	categorical (72)	Neighbourhood inside the city
property_type	categorical (6)	Single Family, Condo, Duplex, Triplex, Townhouse, ...
garage	categorical (Yes/No)	Whether the property has a garage
basement	categorical (Yes/No)	Whether the property has a basement
bedrooms	continuous (discrete count)	Number of bedrooms (1-8)
bathrooms	continuous (discrete count)	Number of bathrooms (1-6)
living_area_sqft	continuous	Living area, square feet
lot_size_sqft	continuous	Lot size, square feet (missing for condos)
year_built	continuous (year)	Year of construction (1900-2023)
renovation_year	continuous (year)	Year of renovation (2005-2025); missing = not renovated
sale_year	continuous (year)	Year of sale (2021-2025)
sale_price	continuous, TARGET	Sale price in dollars

Tasks of the work:

- analyse the quality of the data set, visualise it and find relationships between the variables;
- prepare the data for the linear model, using only parameters computed on the training sample;
- implement the least squares method and gradient descent for the coefficients of a multivariable linear regression;
- test the significance of the coefficients and modify the model;
- perform 5-fold cross-validation for three sets of variables and compare them;
- evaluate the models on test data, compare them with scikit-learn and prepare the model for single-record prediction.

Theoretical Background

Linear regression model

Let n be the number of records and p the number of features. The multivariable linear regression model predicts the target as a weighted sum of the features:

$$\hat{y}_i = w_0 + w_1 x_{i1} + w_2 x_{i2} + \ldots + w_p x_{ip} \quad (1)$$

In matrix form, with a column of ones added to the data matrix X (size n × (p+1)) for the intercept w0, and w = (w0, w1, …, wp)^T:

$$\hat{y} = Xw \quad (2)$$

The coefficients are chosen to minimise the mean squared error (MSE) cost function:

$$J(w) = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 = \frac{1}{n}\|y - Xw\|x^2 \quad (3)$$

Method of least squares (normal equation)

J(w) is a convex quadratic function, so its minimum is where the gradient is zero:

$$\frac{\partial J}{\partial w} = -\frac{2}{n}X^T(y - Xw) = 0 \quad (4)$$

which gives the normal equations $X^TX\,w = X^Ty$ and the closed-form solution:

$$w = (X^TX)^{-1}X^Ty \quad (5)$$

The inverse exists only if the columns of X are linearly independent. When some columns are (almost) collinear, X^TX is singular or badly conditioned. To keep the computation stable the implementation uses the Moore-Penrose pseudo-inverse, $w = (X^TX)^+X^Ty$, which returns the solution with the smallest norm. Computational cost is about $O(np^2 + p^3)$.

Gradient descent

Gradient descent finds the minimum of J(w) iteratively. The gradient of the cost function is:

$$\nabla J(w) = \frac{2}{n}X^T(Xw - y) \quad (6)$$

Starting from w(0) = 0, the weights are updated against the gradient with learning rate α:

$$w^{(t+1)} = w^{(t)} - \alpha\,\nabla J(w^{(t)}) \quad (7)$$

For each coefficient this means:

$$w_j \leftarrow w_j - \alpha\,\frac{2}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)x_{ij} \quad (8)$$

Because J is a quadratic with Hessian $H = (2/n)X^TX$, the iteration converges when $0 < \alpha < 2/\lambda_{max}(H)$, where λ_{max} is the largest eigenvalue of H. The convergence speed depends on the condition number of X^TX, which is why the features must be standardised first. One iteration costs O(np), so the total cost is O(Tnp) for T iterations. Gradient descent only approaches the least squares solution; directions in which the cost function is almost flat (collinear features) converge very slowly.

Data preparation formulas

Standardisation (z-score) uses the mean μ and standard deviation σ of the training sample only:

$$z = \frac{x - \mu_{train}}{\sigma_{train}} \quad (9)$$

Outliers are capped using the interquartile range IQR = Q3 − Q1 of the training sample:

$$x_{capped} = min(\,max(\,x, Q_1 - 1.5 \cdot IQR\,), Q_3 + 1.5 \cdot IQR\,) \quad (10)$$

A categorical variable with k categories is converted into k − 1 binary (dummy) columns; the omitted category is the reference level. Keeping all k columns would make the dummy columns sum to the intercept column (dummy variable trap).

Significance of coefficients

For each coefficient the hypothesis H0: wj = 0 is tested against H1: wj ≠ 0. The residual variance, covariance matrix of the estimates, standard error and t-statistic are:

$$\hat{\sigma}^2 = \frac{RSS}{n-p-1} = \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{n-p-1} \tag{11}$$

$$Cov(\hat{w}) = \hat{\sigma}^2 (X^T X)^{-1} \tag{12}$$

$$SE(\hat{w}_j) = \sqrt{Cov(\hat{w})}\ (diagonal\ element\ j),\ \ t_j = \frac{\hat{w}_j}{SE(\hat{w}_j)} \tag{13}$$

The two-sided p-value is p = 2(1 − Φ(|t|)), where Φ is the standard normal CDF, Φ(t) = ½(1 + erf(t/√2)). Since the number of degrees of freedom (n − p − 1 = 3,882) is large, the t-distribution is practically the normal distribution. A coefficient is significant if p < α = 0.05.

Error metrics and cross-validation

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|,\ \ RMSE = \sqrt{MSE},\ \ R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y}_i)^2} \tag{14}$$

In k-fold cross-validation the training data is split into k parts. Each part is used once as the validation set while the model is fitted on the other k − 1 parts. The mean and standard deviation of the metric over the folds are:

$$m^k = \frac{1}{k}\sum_{f=1}^{k} m_f,\ \ s = \sqrt{\frac{1}{k-1}\sum_{f=1}^{k}(m_f - \bar{m})^2} \tag{15}$$

2. Data Quality Analysis

2.a Summary of the data set

The data set has 5,000 records. Statistics of the continuous variables (cardinality counts distinct non-missing values):

Feature	N	Miss. %	Card.	Min	Max	Q1	Q3	Mean	Median	Std
bedrooms	5000	0.00	8	1	8	2	4	3.49	3	1.36
bathrooms	5000	0.00	6	1	6	2	3	2.33	2	0.97
living_area_sqft	5000	0.00	1670	469	2967	915	1650	1313.69	1273	472.11
lot_size_sqft	5000	24.22	3144	1570	14834	4086	8005	6257.15	5767	2811.43
year_built	5000	0.00	124	1900	2023	1966	2004	1983.07	1985	26.48
renovation_year	5000	68.16	21	2005	2025	2010	2020	2015.05	2015	5.76
sale_price	5000	0.00	803	114000	1425000	352750	584000	480214.00	453000	177125.39
sale_year	5000	0.00	5	2021	2025	2022	2025	2023.41	2024	1.33

Statistics of the categorical variables:

Feature	N	Miss. %	Card.	Mode	Freq.	%	Mode 2	Freq.	%
city	5000	0.00	32	Montréal	919	18.38	Québec	529	10.58
neighborhood	5000	0.00	72	Centre-Ville	625	12.50	West Island	133	2.66
property_type	5000	0.00	6	Single Family	2254	45.08	Condo	1211	24.22
garage	5000	0.00	2	Yes	3476	69.52	No	1524	30.48
basement	5000	0.00	2	Yes	3032	60.64	No	1968	39.36

Main observations. The target sale_price has a mean of 480,214 and a median of 453,000 (standard deviation 177,125), so it is moderately right-skewed. Only two variables have missing values: lot_size_sqft (24.22 %, 1,211 records) and renovation_year (68.16 %, 3,408 records). The number of missing lot sizes equals exactly the number of condos (1,211), which shows that the values are missing systematically. Garage and basement are present in 69.5 % and 60.6 % of the properties.

2.b Histograms and distributions

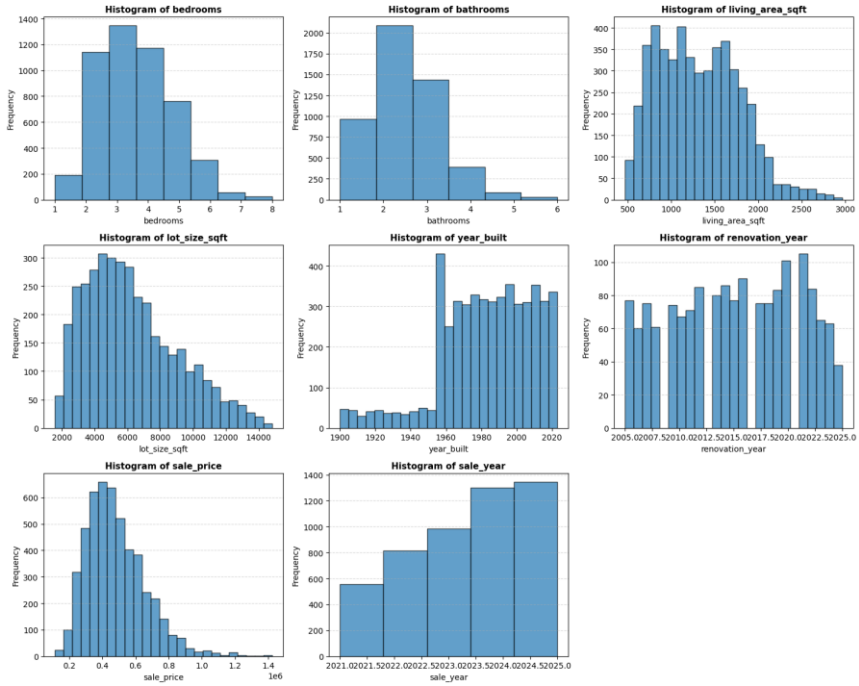


Figure 1. Histograms of the continuous variables

- bedrooms: discrete, unimodal, slightly right-skewed distribution with the mode at 3 bedrooms and rare values of 7-8.
- bathrooms: discrete, right-skewed, the mode is 2 bathrooms; frequency drops sharply after 3.
- living_area_sqft: right-skewed, broad and slightly multi-peaked between about 600 and 1,900 sq ft, with a long tail of large properties up to 2,967 sq ft (mean 1,314 > median 1,273).
- lot_size_sqft: clearly right-skewed, unimodal with the peak at 4,000-5,000 sq ft and a tail to 14,800 sq ft (condos are not shown, their lot size is missing).
- year_built: not a normal distribution; a low, nearly flat level of about 40 properties per bin in 1900-1955 followed by a jump after 1955 to a roughly uniform level of about 300 per bin until 2023.
- renovation_year: approximately uniform between 2005 and 2025 among the 1,592 renovated properties.
- sale_price (target): right-skewed, close to log-normal, with the mode near 400,000 and a long tail to 1.43 million. This suggests that a logarithmic transformation of the target could improve a linear model.
- sale_year: discrete (2021-2025); the number of sales grows from about 550 in 2021 to about 1,350 in 2025.

2.c Data quality issues and how they are handled

1. Missing values.

- **renovation_year (68.16 % missing).** A missing year means that the property was never renovated, not that the information was lost. Handling: binary feature is_renovated (1 if a year exists, 0 otherwise) and renovation_year_clean in which missing values are replaced by 0. The model can then use both the fact and the year of renovation.
- **lot_size_sqft (24.22 % missing).** All missing values belong to condos, whose land is shared. Handling: imputation with the median of the training sample (robust to the skewed distribution). Deleting these rows would remove a quarter of the data, and the dummy variable property_type_Condo lets the model correct for the systematic difference.

2. Outliers. According to the 1.5·IQR rule, living_area_sqft has 19 values above 2,752.5 sq ft and lot_size_sqft has 23 values above 13,883.5 sq ft. Least squares squares the residuals, so a few extreme points can strongly shift the coefficients. The values are therefore capped (winsorised) at the IQR limits computed on the training sample instead of being deleted, which keeps all records. The target sale_price (up to 1.43 million) was not modified because these are plausible real sales.

3. Redundant and derived variables. house_age = sale_year − year_built is an exact linear combination of two other features. It was created as a derived variable but excluded from the final model, otherwise the matrix X^TX would be singular.

4. High-cardinality categorical variables. city (32) and neighborhood (72) produce 31 + 71 dummy columns. Grouping rare categories into "Other" was considered, but in the final pipeline all categories were encoded. As shown in section 3.b-3.c, this has a side effect: for cities with a single neighborhood the city dummy and the neighborhood dummy are identical columns.

5. Skewed target and non-linearity. The right-skewed target and the widening spread of the price with the living area (Figure 2) indicate heteroscedasticity; this is discussed in the error analysis (3.f).

2.d Relationships between variables

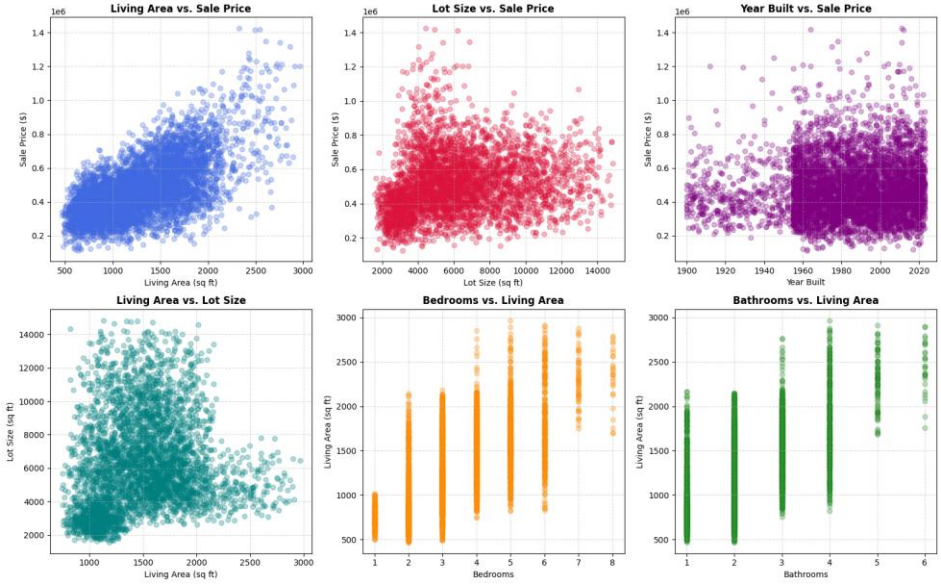


Figure 2. Scatter plots of selected variables

- Living area vs. sale price: strong positive, roughly linear dependence; the spread of prices increases for larger houses.
- Lot size vs. sale price: weak positive dependence that flattens for large lots.
- Year built vs. sale price: no visible pattern; old and new houses have a similar price range.
- Living area vs. lot size: no relationship, i.e. the lot size carries information independent of the house size.
- Bedrooms / bathrooms vs. living area: step-wise increase of the living area with the number of rooms, so these three variables are partly redundant.

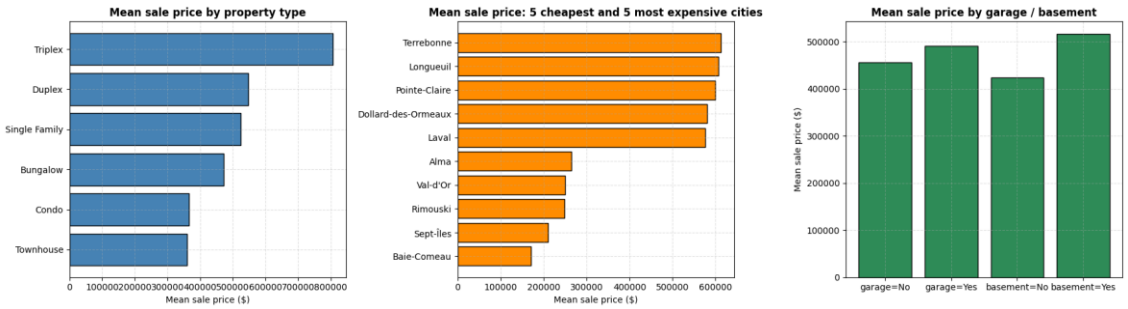


Figure 3. Bar plots of mean sale_price by property type, city and garage/basement

2.e Correlation matrix

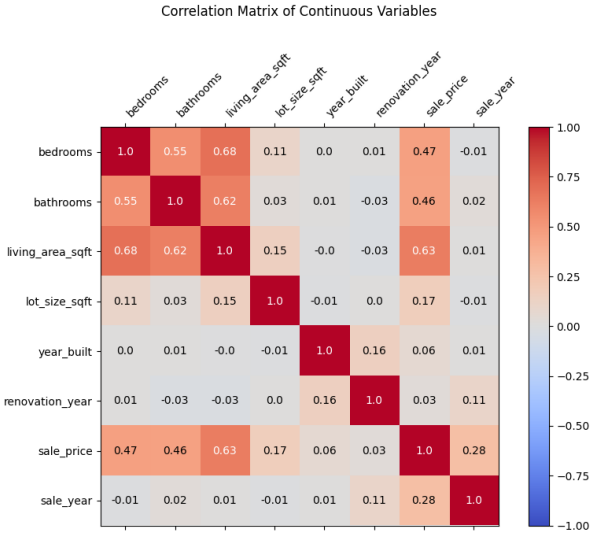


Figure 3. Correlation matrix of continuous variables

Correlation with the target: the strongest linear relationship is with living_area_sqft (r = 0.63), followed by bedrooms (0.47), bathrooms (0.46), sale_year (0.28) and lot_size_sqft (0.17). year_built (0.06) and renovation_year (0.03) are almost uncorrelated with the price. The positive sale_year correlation shows that prices grew over the observed period.

Correlation between predictors: living_area_sqft is strongly correlated with bedrooms (0.68) and bathrooms (0.62); bedrooms and bathrooms are correlated with 0.55. These three variables carry overlapping information (multicollinearity), which inflates the variance of their coefficients. The correlations are not large enough ($|r| < 0.7$) to make the model singular, so all of them were kept. The remaining pairs are below 0.16. The matrix describes only linear relations between numerical variables; the effect of city, property type, garage and basement is not visible in it.

2.f Data preparation for the regression model

All parameters were calculated only on the training sample and then applied unchanged to the test sample.

- **Split:** random 80/20 split (seed 42): 4,000 training and 1,000 test records, performed before any other processing.
- **Derived variables:** is_renovated, renovation_year_clean (0 if not renovated), house_age = sale_year − year_built (later removed because of exact collinearity).
- **Missing values:** lot_size_sqft filled with the training median.
- **Outliers:** living_area_sqft and lot_size_sqft capped with the training Q1 − 1.5·IQR and Q3 + 1.5·IQR limits (formula 10).
- **Categorical variables:** one-hot (dummy) encoding of city, neighborhood, property_type, garage and basement, dropping the first (alphabetical) category of each to avoid the dummy variable trap.
- **Normalisation:** z-score (formula 9) of the numerical variables bedrooms, bathrooms, living_area_sqft, lot_size_sqft, year_built, renovation_year_clean and sale_year with the training mean and standard deviation. Dummy variables are left as 0/1.

3. Linear Regression Model

3.a Implementation of the methods

Both methods were implemented with NumPy only, following formulas (5) and (6)-(7). A column of ones is added to X for the intercept.

```
def fit_linear_regression_ols(X, y):
    X_b = np.hstack([np.ones((X.shape[0], 1)), X])
    return np.linalg.pinv(X_b.T @ X_b) @ X_b.T @ y          # w = (X^T X)^+ X^T y

def fit_linear_regression_gd(X, y, learning_rate=0.01, iterations=1000):
    n = X.shape[0]
    X_b = np.hstack([np.ones((n, 1)), X])
    w = np.zeros((X_b.shape[1], 1))
    for _ in range(iterations):
        gradient = (2 / n) * (X_b.T @ (X_b @ w - y))        # grad J(w)
        w = w - learning_rate * gradient                     # w <- w - alpha * grad
    return w
```

Batch gradient descent was used (no modification), with zero initial weights, learning rate $\alpha = 0.05$ and 2,000 iterations. A relatively large α is possible because the numerical features are standardised.

3.b Coefficients of both methods

Table 1. Coefficients of the least squares (OLS) and gradient descent (GD) models (selected rows; all 118 are in the notebook)

Variable	OLS	Gradient descent	|Difference|
Intercept (w0)	752,252	366,828	385,424
bedrooms	9,363	8,018	1,345
bathrooms	10,226	8,014	2,212
living_area_sqft	93,059	89,135	3,924
lot_size_sqft	11,375	11,754	379
year_built	12,365	12,548	183
sale_year	49,975	50,185	210
is_renovated	-1,059,197	100,499	1,159,696
renovation_year_clean	506,531	-33,512	540,043
city_Baie-Comeau	-93,425	-87,810	5,615
neighborhood_Baie-Comeau Centre	-93,425	-87,810	5,615
city_Blainville	26,844	16,068	10,776
neighborhood_Blainville Centre	26,844	16,068	10,776
city_Longueuil	170,127	153,340	16,787
city_Montréal	148,838	125,439	23,399
property_type_Condo	-14,176	15,891	30,067
property_type_Duplex	-57,377	-14,116	43,261
property_type_Single Family	-46,730	-9,075	37,655
property_type_Townhouse	-88,661	-55,149	33,512
property_type_Triplex	49,747	104,495	54,748
garage_Yes	13,476	14,190	714
basement_Yes	8,966	11,277	2,311

Comparison. For the main numerical variables the two methods agree well: living_area_sqft 93,059 (OLS) vs. 89,135 (GD), year_built 12,365 vs. 12,548, sale_year 49,975 vs. 50,185, lot_size_sqft 11,375 vs. 11,754. Since the variables are standardised, a coefficient is the price change for a one-standard-deviation change of the feature; the living area is the most influential numerical variable. Part of the small differences is caused by the finite number of iterations: gradient descent stops before the exact minimum.

Large differences appear for the intercept (752,252 vs. 366,828), is_renovated (−1,059,197 vs. 100,499), renovation_year_clean (506,531 vs. −33,512) and property type dummies. There are two reasons, both related to multicollinearity of the design matrix:

- **is_renovated and renovation_year_clean are almost perfectly dependent** (renovation_year_clean is 0 for all non-renovated houses and about 2015 for the renovated ones). The OLS standard errors of these two coefficients (≈654,000 and ≈304,000, see Table 2) are hundreds of times larger than for the other numerical variables (1,000-2,000), which is a typical symptom of near-collinearity. Their huge opposite-sign coefficients almost cancel each other, so the predictions are not affected.
- **Duplicated dummy columns.** Several cities have only one neighborhood, so the dummy columns are identical, e.g. city_Baie-Comeau and neighborhood_Baie-Comeau Centre (both −93,425), city_Blainville and neighborhood_Blainville Centre (both 26,844). The matrix is rank-deficient; the pseudo-inverse splits the effect between the identical columns, and the intercept and the reference-category effects (property type) absorb the rest.

Gradient descent started from zero and, after a limited number of iterations, found a different but also valid point in this flat valley of the cost function. The individual coefficients of collinear variables are therefore not uniquely interpretable, but the fitted values of the model are practically the same.

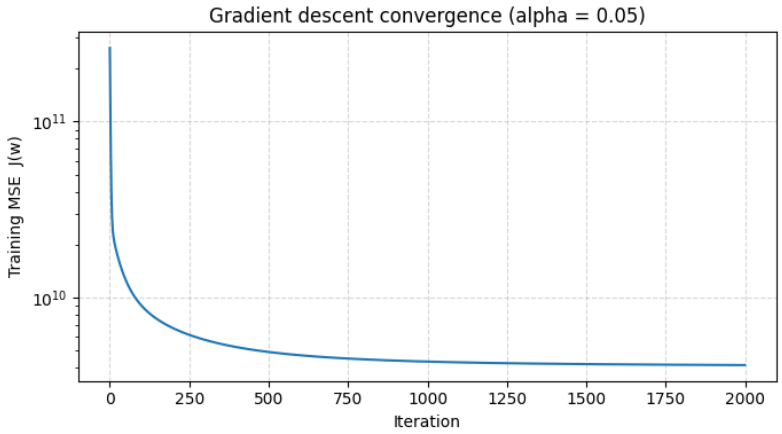


Figure 3. Gradient-descent convergence

3.c Significance of the coefficients

For every coefficient the t-statistic and p-value were calculated with formulas (11)-(13) on the training data (α = 0.05).

Table 2. Selected results of the significance test (OLS model with all 117 variables)

Variable	Coefficient	Std. error	t	p-value	Significant
sale_year	49,975	1,037	48.19	< 0.001	yes
living_area_sqft	93,059	2,118	43.94	< 0.001	yes
city_Longueuil	170,127	8,760	19.42	< 0.001	yes
year_built	12,365	1,132	10.93	< 0.001	yes
lot_size_sqft	11,375	1,471	7.74	< 0.001	yes
bathrooms	10,226	1,518	6.74	< 0.001	yes
bedrooms	9,363	1,666	5.62	< 0.001	yes
garage_Yes	13,476	2,328	5.79	< 0.001	yes
basement_Yes	8,966	2,980	3.01	0.0026	yes

Variable	Coefficient	Std. error	t	p-value	Significant
property_type_Condo	−14,176	7,259	−1.95	0.051	no
renovation_year_clean	506,531	304,448	1.66	0.096	no
is_renovated	−1,059,197	653,699	−1.62	0.105	no
city_Québec	−3,089	9,692	−0.32	0.750	no

Result: 73 of the 117 variables are significant and 44 are not: is_renovated, renovation_year_clean, property_type_Condo, four city dummies (Lévis, Québec, Rouyn-Noranda, Sherbrooke) and 37 neighborhood dummies. The most significant variables (largest |t|) are sale_year, living_area_sqft and city_Longueuil. All the basic numerical variables, garage and basement are significant.

Modification of the model: the 44 non-significant variables were removed and the model was refitted with the least squares method (73 variables). Two remarks must be made. First, removing a single dummy variable merges that category with the reference category, so this reduction changes the meaning of the remaining categories. Second, for duplicated city/neighborhood columns the standard errors are not reliable because the matrix is rank-deficient, so their p-values should be interpreted with caution.

3.d Cross-validation of the model

5-fold cross-validation (shuffled with seed 42, 800 records per fold) was performed on the training data with the least squares method for the model with significant variables. Error metrics: MSE and RMSE.

Table 3. 5-fold cross-validation, significant variables, least squares

Fold	MSE	RMSE
Fold 1	4,738,161,000	68,834
Fold 2	4,370,986,000	66,113
Fold 3	4,496,116,000	67,053
Fold 4	4,311,501,000	65,662
Fold 5	4,291,420,000	65,509
Mean	4,441,637,000	66,634

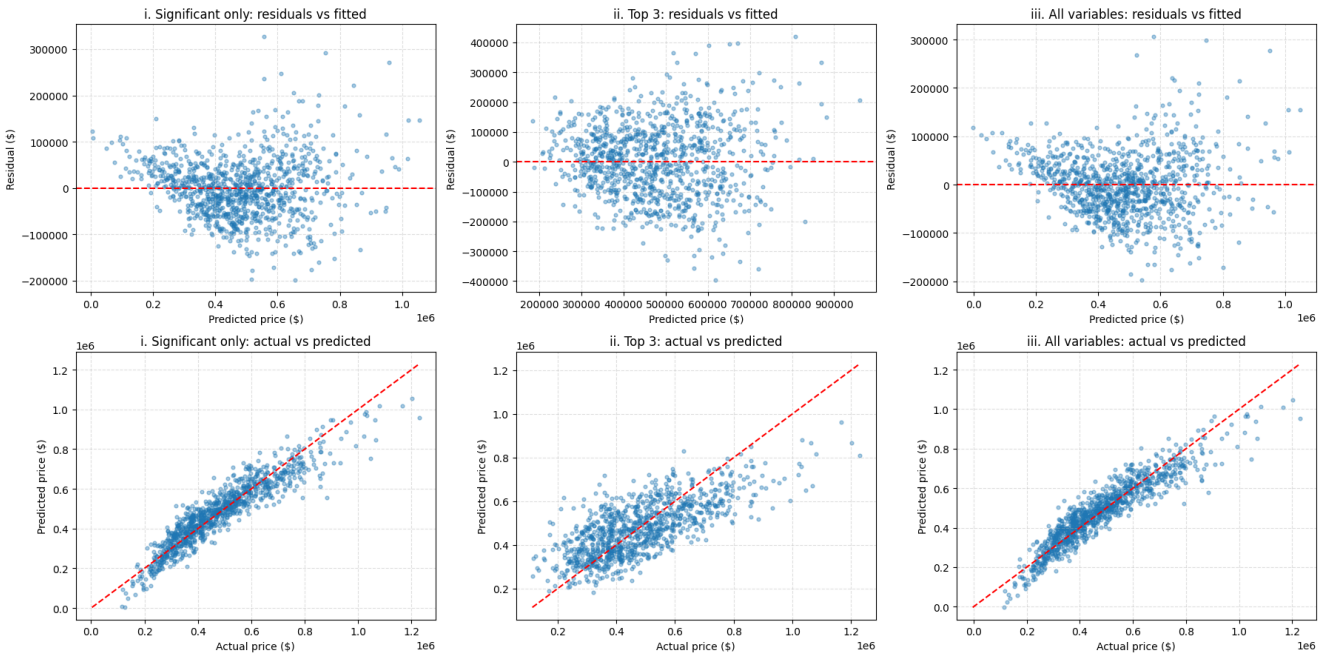
The RMSE varies only between 65.5 and 68.8 thousand, so the model is stable with respect to the choice of the training subset. An average error of about 66,600 is about 14 % of the mean price.

3.e Cross-validation for three sets of variables

The 5-fold cross-validation was repeated for (i) only significant variables (73), (ii) only the 3 most significant variables, (iii) all 117 variables. The mean and standard deviation of the RMSE over the folds are given below.

Table 4. Mean and standard deviation of RMSE in 5-fold cross-validation

Model	Variables	Mean RMSE	Std of RMSE
i. Significant only	73	66,634	1,369
ii. 3 most significant	3	123443	2320
iii. All variables	117	65,594	1,251



Comments. The model with all variables (iii) has both the lowest mean error (65,594) and the lowest dispersion (1,251). The model with significant variables only (i) is about 1,000 (1.6 %) worse and slightly more dispersed. The difference is comparable to the fold-to-fold standard deviation, so it is small, but it shows that removing the non-significant dummy variables did not improve generalisation; they still carry a little information, and removing them only simplifies the model (73 instead of 117 variables). The model with 3 variables (ii) is expected to have a considerably higher error and the largest dispersion, because most of the information (property type, other cities, rooms, lot size) is missing.

3.f Evaluation on the test data

The models were fitted on the whole training sample and evaluated on the 1,000 unseen test records.

Table 5. Error metrics on the test data (least squares)

Model	MAE ($)	MSE	RMSE ($)	R^2
i. Significant only (73)	49,345	$4.172 \cdot 10^9$	64,591	0.8625
ii. 3 most significant	97,074	$14{,}61 \cdot 10^9$	120905	0.51
iii. All variables (117)	49,161	$4.108 \cdot 10^9$	64,093	0.8646

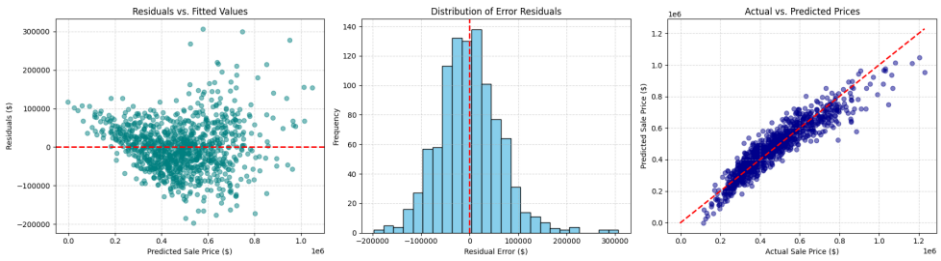


Figure 4. Error analysis of the model with all variables on the test data

Both full models explain about 86 % of the variance of the price. The test RMSE (64,093 and 64,591) is slightly lower than the cross-validation RMSE (65,594 and 66,634), which means that the model does not overfit. The MAE of about 49,000 corresponds to roughly 10 % of the mean price. As in cross-validation, the model with all variables is marginally better.

Error analysis (Figure 4): the residuals are centred around zero and their histogram is approximately symmetric and bell-shaped (from about −200,000 to +300,000), so the model is unbiased on average. However, the residuals vs. fitted plot is curved (positive residuals for the lowest and the highest predictions) and the spread grows with the predicted price (funnel shape): the relationship is not perfectly linear and the variance of errors is not constant (heteroscedasticity). In the actual vs. predicted plot the points for expensive houses (> 0.8 million) lie below the

diagonal, i.e. the model underestimates the highest prices. A logarithmic transformation of the target, or interaction terms (living area × city), would probably reduce these effects.

3.g Comparison with scikit-learn

The same model (all variables) was built with sklearn.linear_model.LinearRegression and compared with the own least squares implementation on the test data.

Table 6. Own implementation vs. scikit-learn (test data)

Metric	Own implementation	scikit-learn	|Difference|
MAE ($)	49,161.14	49,161.14	$1.4 \cdot 10^{-7}$
MSE	4,107,915,000	4,107,915,000	0.024
RMSE ($)	64,093.02	64,093.02	$1.9 \cdot 10^{-7}$
R^2	0.864628	0.864628	$7.9 \cdot 10^{-13}$
Intercept w0	752,252	791,569	39,317

The predictions and all error metrics are identical up to numerical precision, which confirms that the own implementation of the least squares method is correct. Only the intercept differs. This is explained by the rank deficiency of the design matrix (identical dummy columns and near-collinear renovation features): the solution of the normal equations is then not unique, and the pseudo-inverse of the own implementation and the solver of scikit-learn (which first centres the data) select different, equally good, solutions with the same fitted values.

3.h Using the model with a new record

The final model (least squares, all variables) is wrapped in two functions. preprocess_record() accepts one record in the original format (a dictionary with the raw values of the 12 features) and applies exactly the same transformations as in training with the saved training parameters: derived features (is_renovated, renovation_year_clean), median imputation and IQR capping of the lot size, capping of the living area, standardisation with the training mean and standard deviation, and one-hot encoding. predict_price() multiplies the resulting vector with the coefficients and returns the price.

```
new_house = {
    'city': 'Laval', 'neighborhood': 'Chomedey', 'property_type': 'Single Family',
    'bedrooms': 3, 'bathrooms': 2, 'living_area_sqft': 1500, 'lot_size_sqft': 5500,
    'year_built': 1995, 'renovation_year': None, 'garage': 'Yes', 'basement': 'Yes',
    'sale_year': 2025}
print(predict_price(new_house))
```

Predicted price (single record) : 223,197.95
Predicted price (batch pipeline) : 223,197.95
Actual price   : 259,000.00

New house predicted price: 644,943 $

Conclusions

A multivariable linear regression model was implemented from scratch using Least Squares and Gradient Descent. The analysis showed that living area, sale year, location and property type are important predictors of house prices.

The model achieved good predictive performance, with \(R^2=0.865\), RMSE ≈ $64,100 and MAE ≈ $49,200 on the test data. Cross-validation showed similar results, suggesting that the model generalizes reasonably well to unseen data. The from-scratch implementation also produced results consistent with scikit-learn.

Some coefficients were affected by multicollinearity, and the residual analysis indicated possible non-linearity and increasing variance. Future improvements could include removing redundant variables, transforming the target variable, or using regularization.